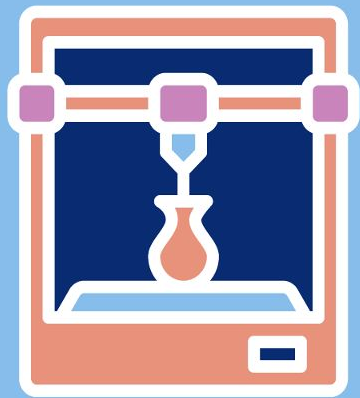




ACTION LAB GRAD



Kaixin Zeng
Project Portfolio

Digital Prototyping

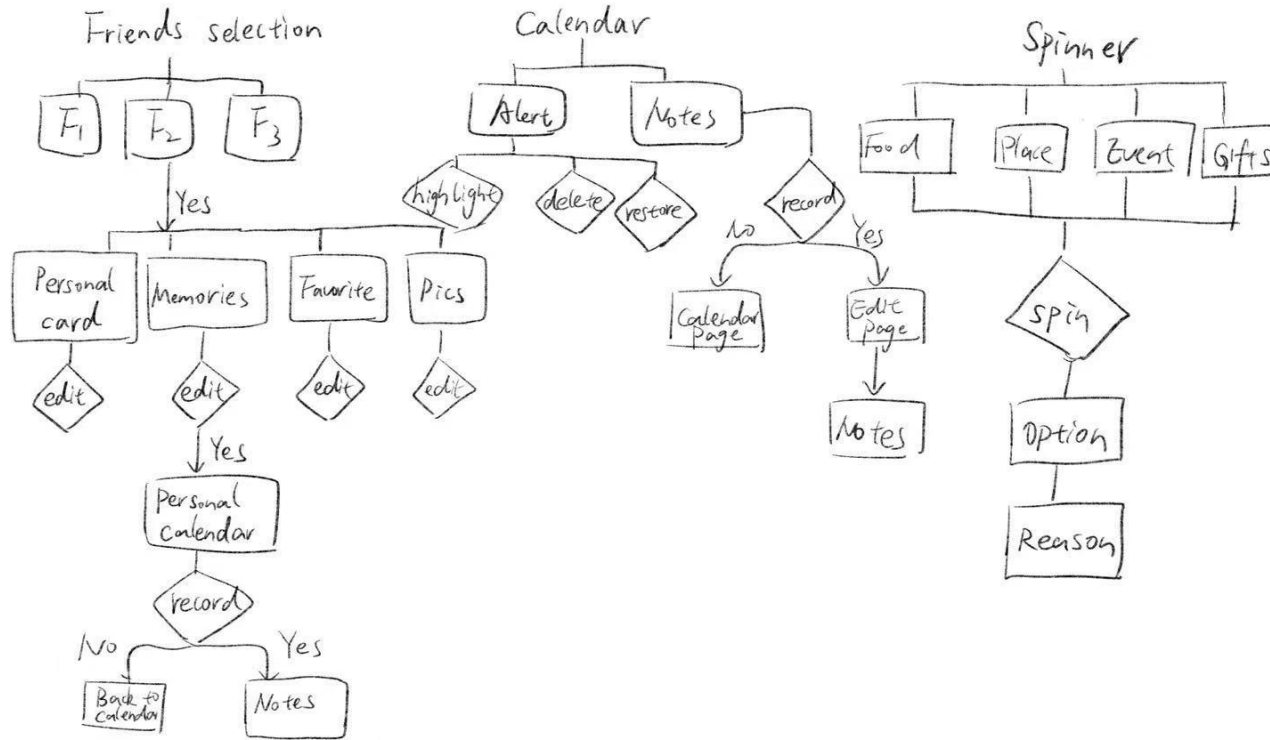
Designing a digital interface

using sketching, paper prototyping,
figma, & graphic design.

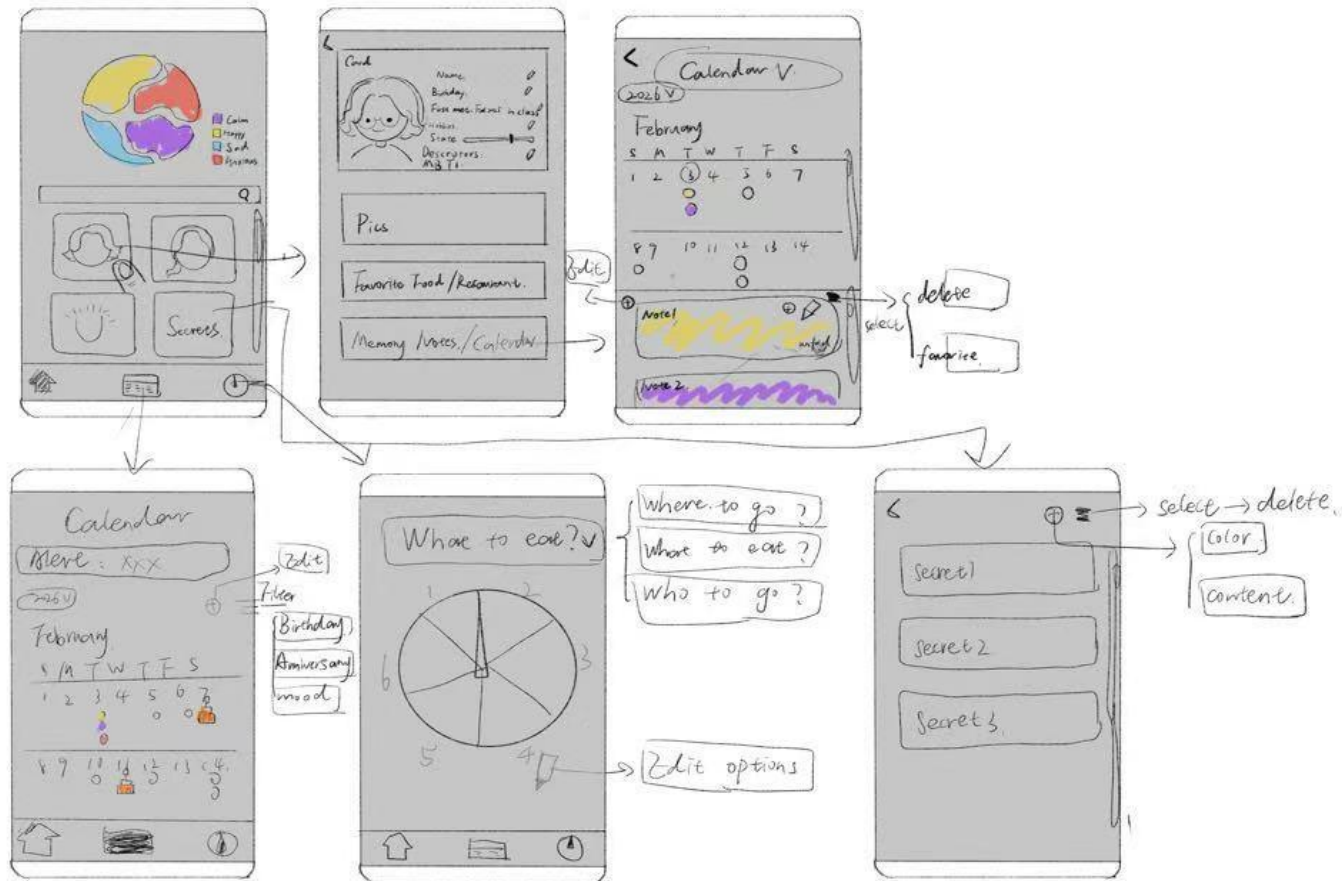
Friend Notes

A quiet place to remember the people
who matter

Wireframe Sketch

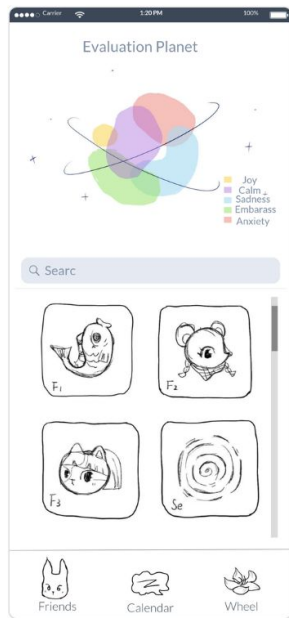


Initial Block Sketch



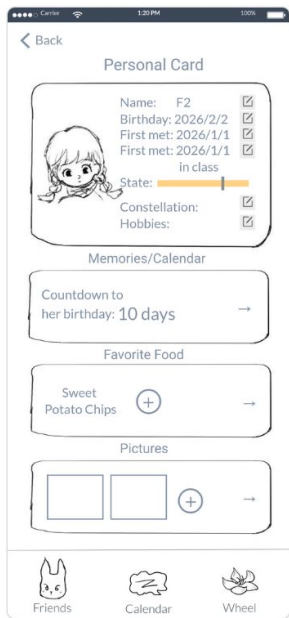
Low Resolution Prototype 1:

MARVEL DESIGN



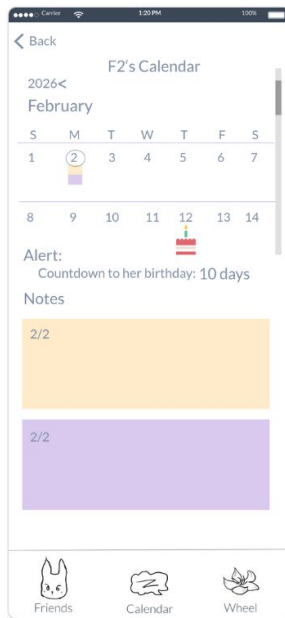
Updated 2nd Feb @ 19:20 PM

Image 1



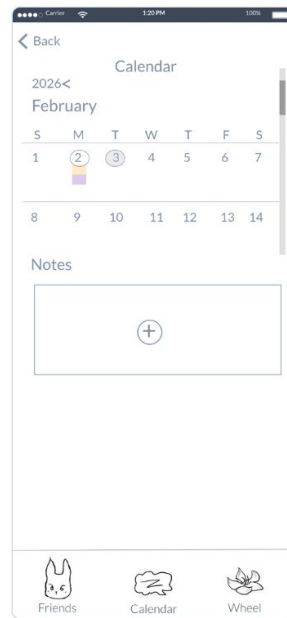
Updated 2nd Feb @ 18:51 PM

Image 2



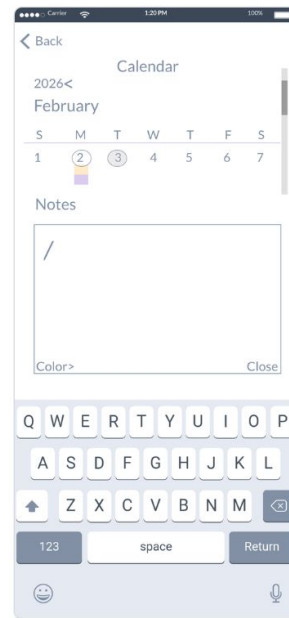
Updated 2nd Feb @ 18:57 PM

Image 3



Updated 2nd Feb @ 18:56 PM

Image 4

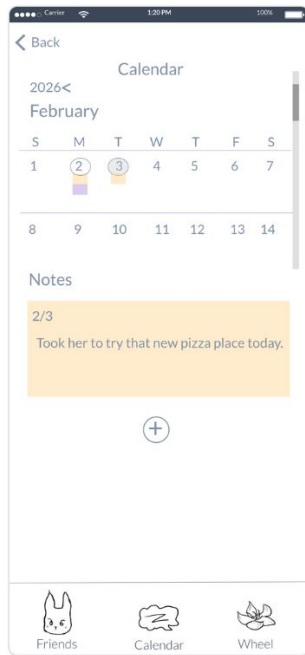


Updated 2nd Feb @ 18:50 PM

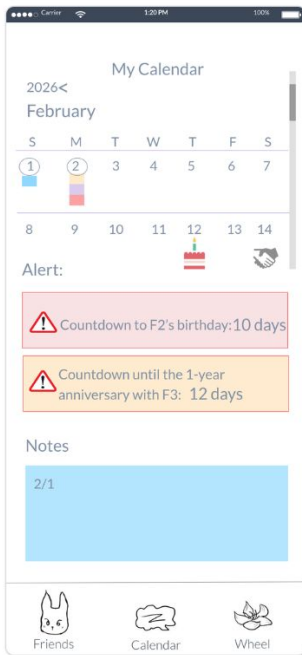
Image 5

Low Resolution Prototype 1:

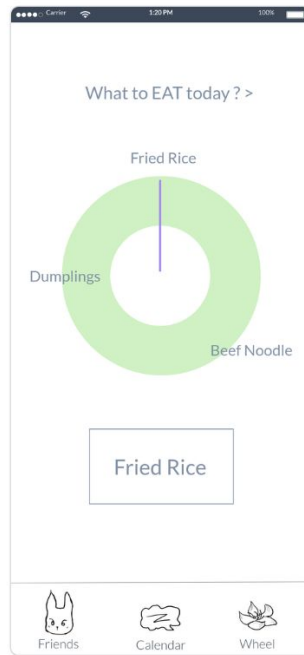
MARVEL DESIGN



Updated 2nd Feb @ 19:23 PM



Updated 2nd Feb @ 19:38 PM



Updated 2nd Feb @ 19:38 PM

<https://marvelapp.com/prototype/d1ec0i5>

Image 6

Image 7

Image 8

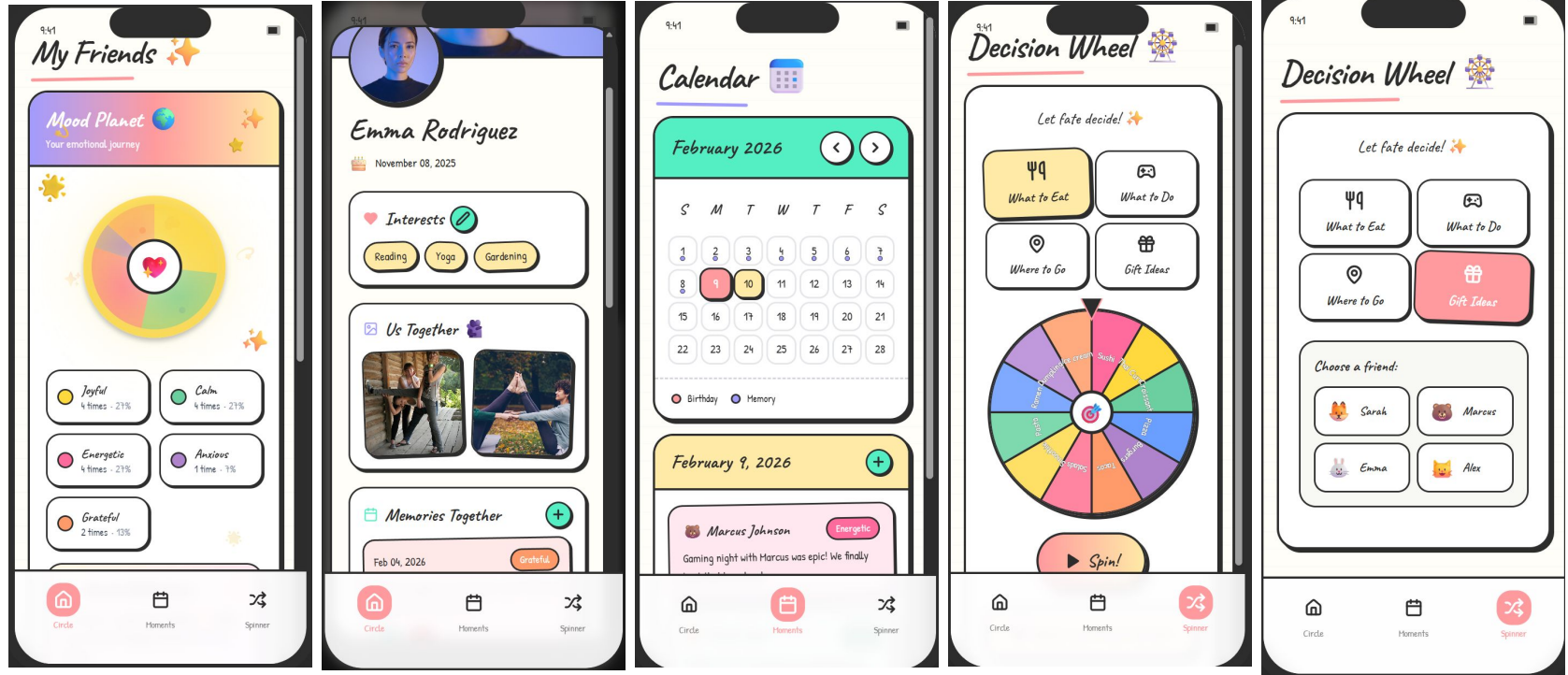
Low Resolution Prototype 1:

MARVEL TESTING

1. Clearer, single-path user flows
2. A more consistent button and interaction structure
3. More engaging and practically useful features

Medium Resolution Prototype 2:

FIGMA DESIGN

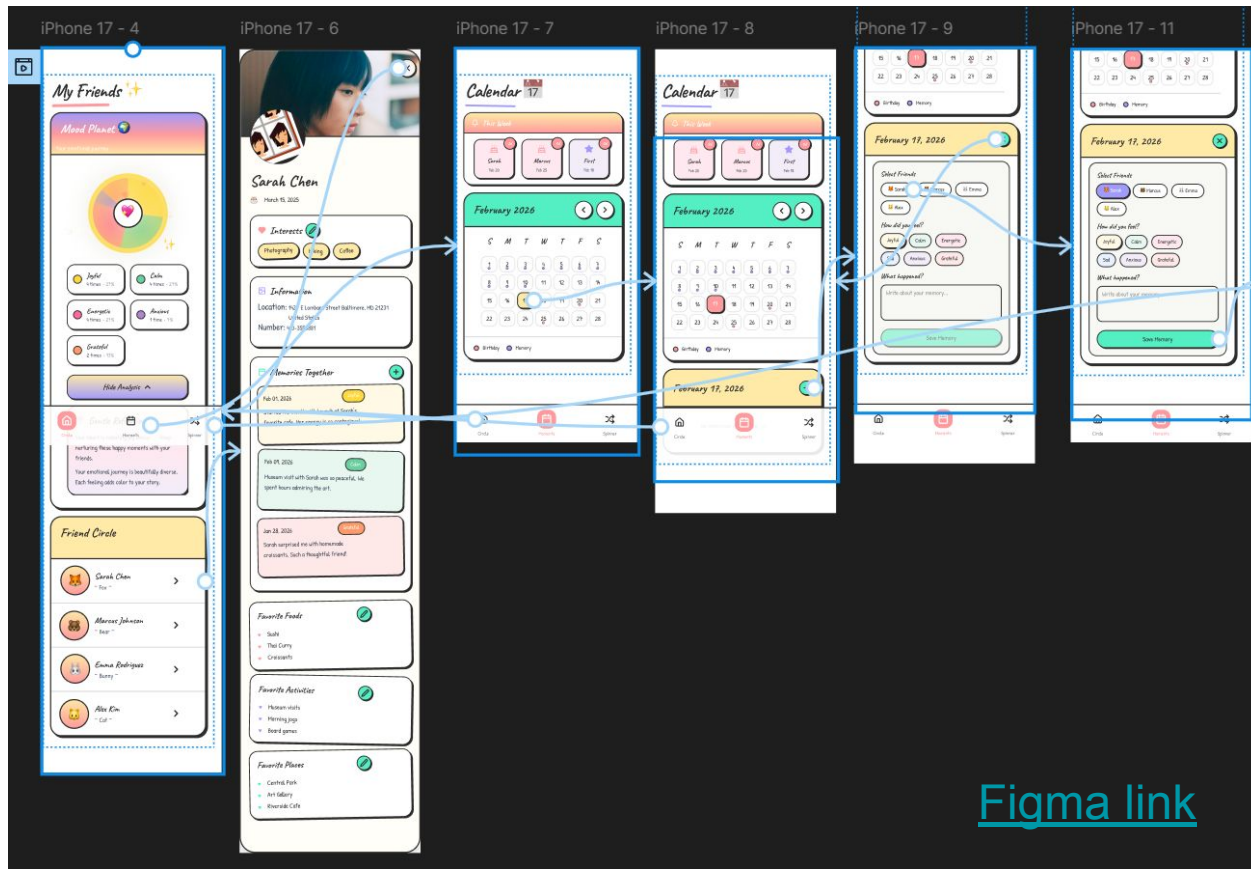


Medium Resolution Prototype 2:

FIGMA TESTING

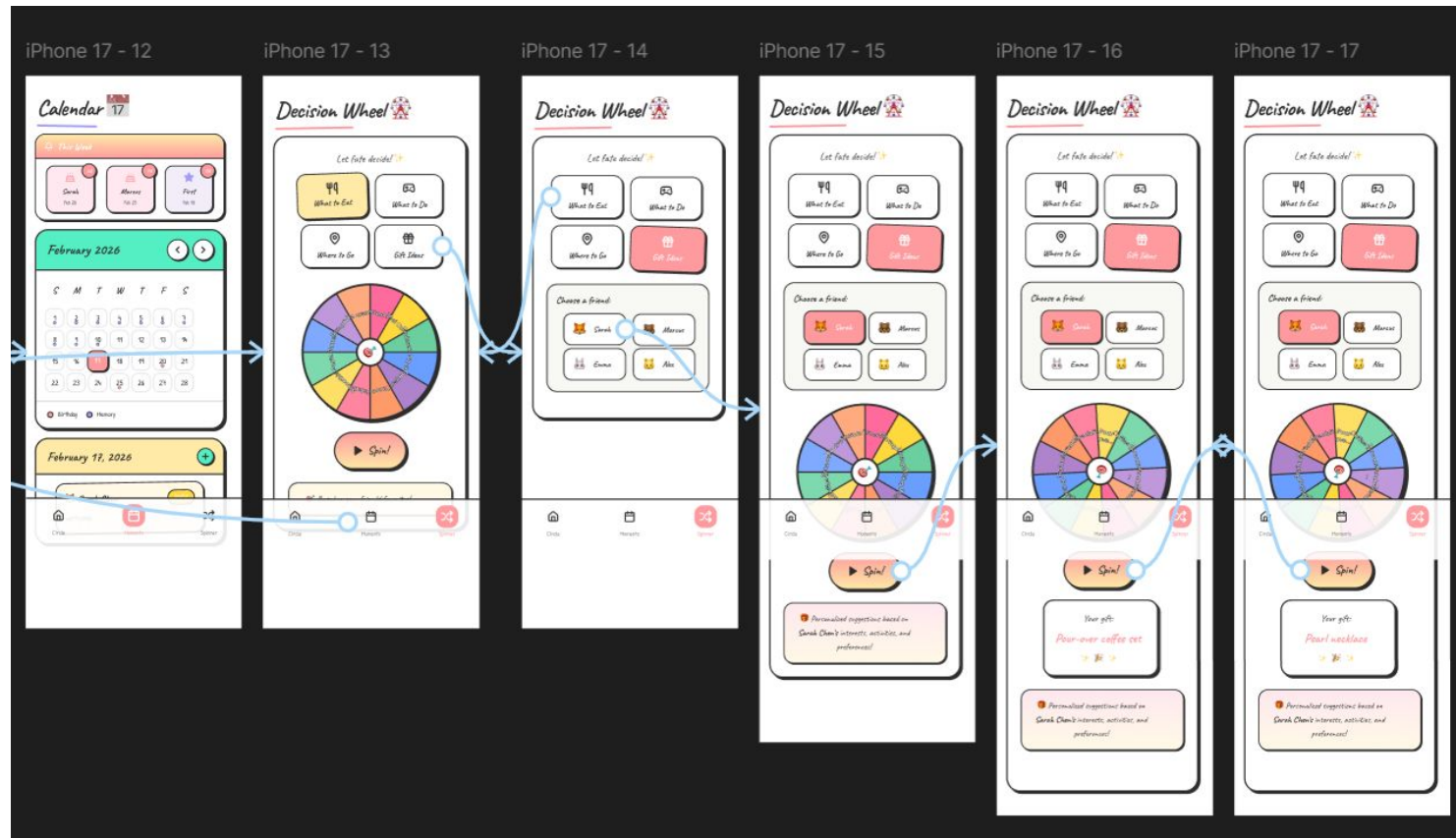
1. Add a birthday reminder feature.
2. Allow users to store friends' addresses and phone numbers.
3. Use clearer and more intuitive section labels.

Final Product



[Figma link](#)

Final Product



Physical Prototyping

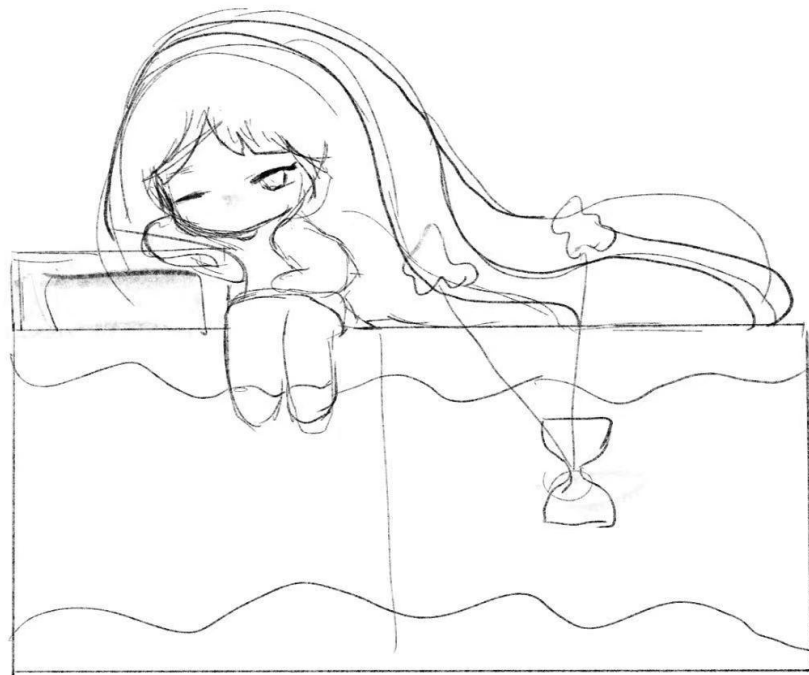
Designing a physical product

using sketching, computer-aided design,
laser cutting, 3D printing, & electronics.

Tiny Garden Clock

A handmade digital clock that combines
real-time display with a playful,
garden-inspired decorative scene.

Initial Sketch



Cardboard Prototype



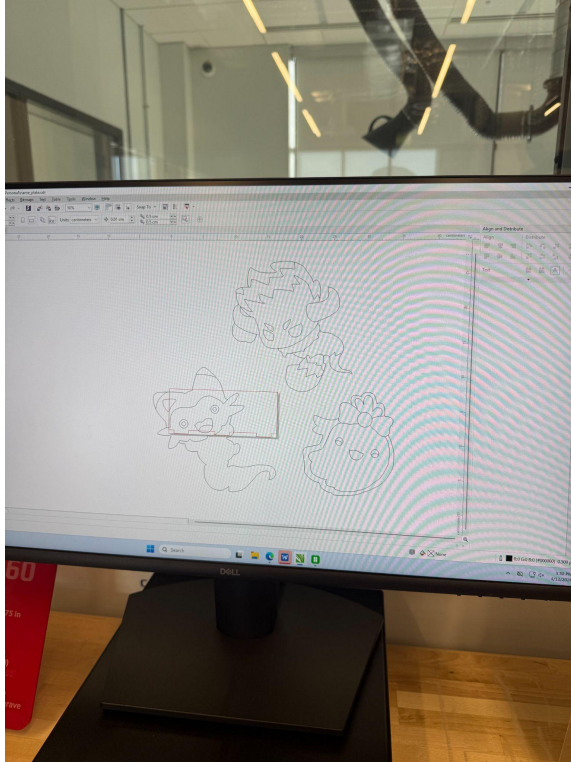
Cardboard is a great tool for testing design effectiveness and structural stability.

Laser Cutting Prototype



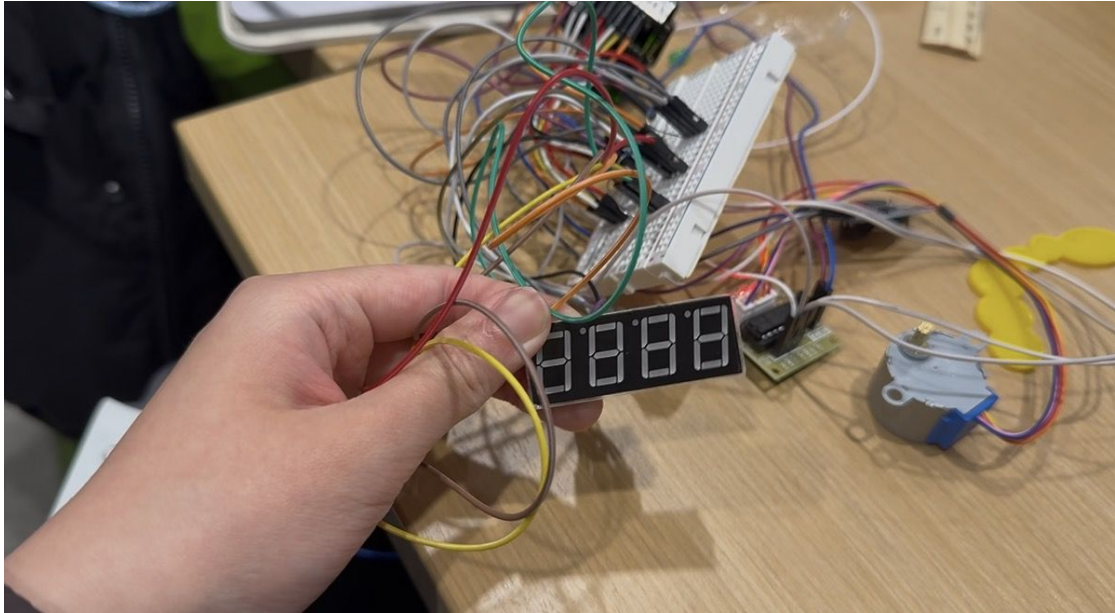
I need to further familiarize with the design software to create designs that clearly distinguish between areas for full cutting and engraved patterns.

Laser Cutting Prototype: Iteration



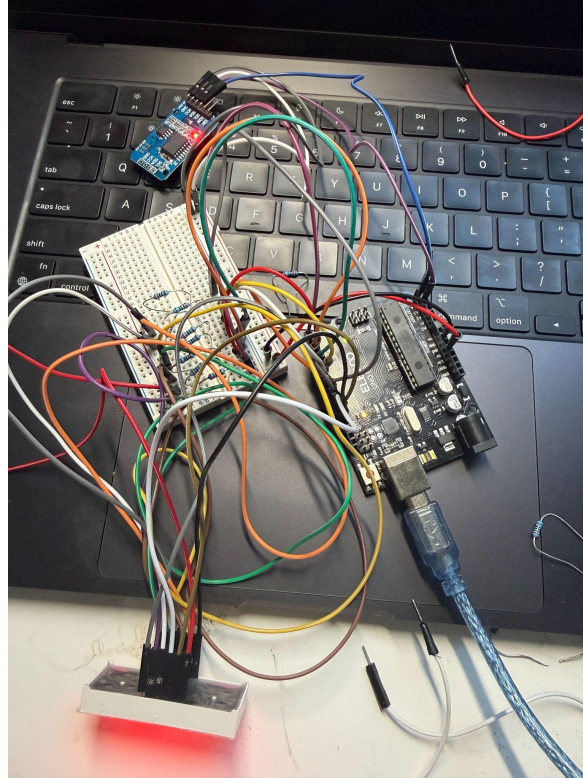
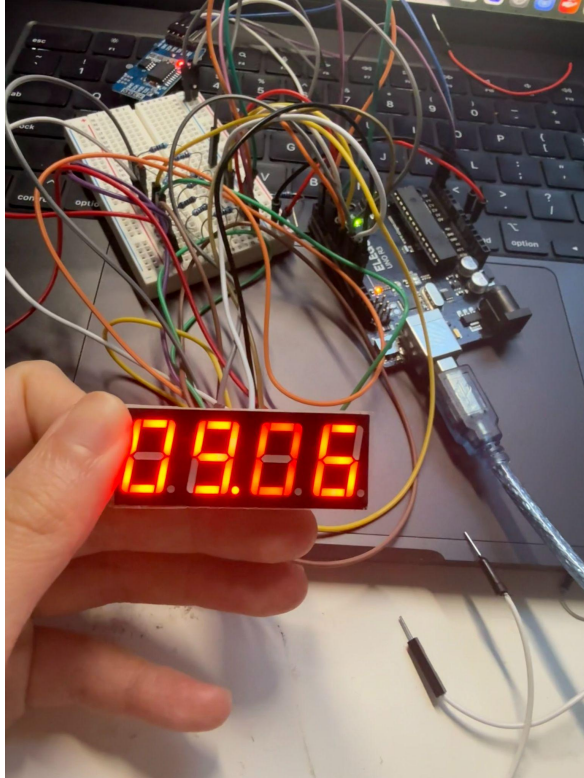
1. I used design software to create the laser cutting drawings for making the figures.
2. I need to plan which parts should be printed in each material color.

Electronics Prototype



My Arduino programming is not yet complete and requires further refinement to function correctly.

Electronics Prototype: Iteration



My 4-digit LED display can finally show the real time properly.

3D Printing Prototype



I need to adjust the model dimensions and continue refining the gear design to achieve the intended sketch outcome.

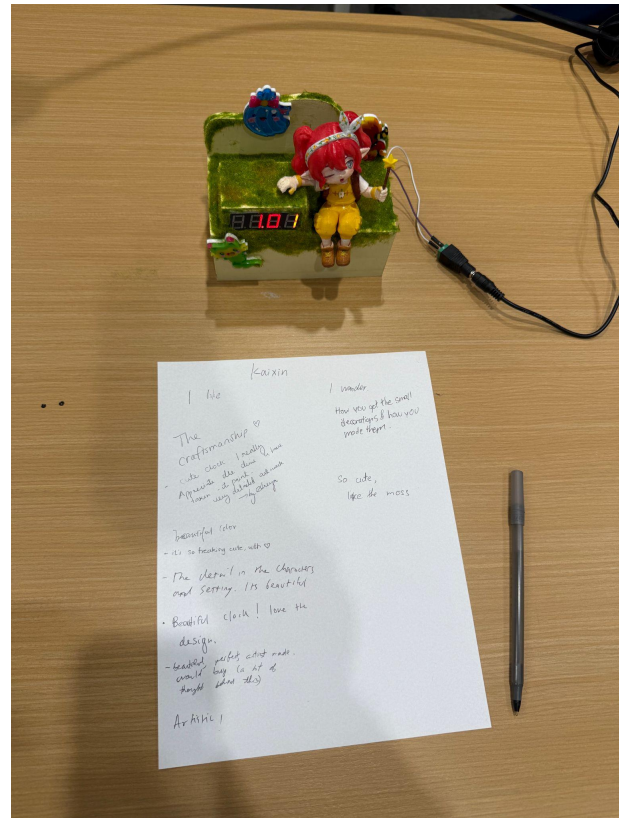
3-D Printing Prototype: Iteration

<https://a360.co/4vwd9nF>

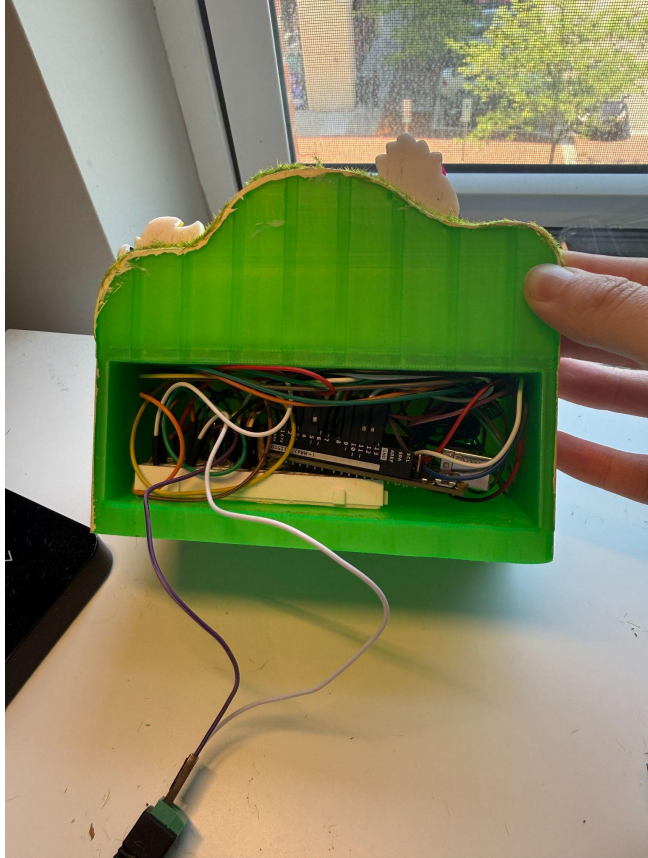


1. I printed the main body of the clock again, but the color was slightly different from what I expected.
2. I could continue trying other colors.

Final Product



Final Product



Management Prototyping

*Designing systems,
processes, or products
for project efficacy and efficiency*

Identified Needs

FOCUS AREA: EP course team

1. Communication Structure Needs

Need for clear asynchronous communication protocols

2. Workflow Needs

Need for predictable decision-making timelines

Need for minimized bottlenecks caused by delayed replies

3. Team Dynamics Needs

Need for improved team cohesion despite physical distance

Need for reduced frustration caused by communication lag

Brainstormed Solutions

NEED: Address un-asynchronous communication's problems

My top 6 ideas to address the need:

- **Confirm the timeline for each task early to prevent unexpected delays.**
- **Agree in advance on how many hours each message should be answered within.**
- Turn on special notifications for this project.
- **Use shared files to track team members' progress in real time and remind each other when needed.**
- Establish a clear accountability and follow-up system.
- Hold regular meetings to align information and update each member's status.

AI's top 6 ideas to address the need:

- Set clear roles and ownership from the start.
- **Break the project into small tasks with specific deadlines.**
- Use one main communication channel to avoid missed information.
- Create a shared progress tracker that everyone can see.
- **Schedule short weekly check-ins to catch problems early.**
- **Set team rules for response time, meeting attendance, and work quality.**

- Highlight your top 3 (from both lists combined)

Align

A System for Smarter Group Work

Initial Sketch

[Onboarding Information Page – Figma Make](#)

1. automatically forms teams based on people' s schedules, skills, and preferences, so it' s easier to find a common time and have a balanced group.
2. helps assign tasks **by suggesting who should do what**
3. tracks **each person' s contribution** to **visualize** and make the work more fair and keep everyone involved.

Management Prototype

By creating a Figma page, I want to help online courses' teams to generate a team that best fit their available times and skills.

Tell us about your availability & skills

Help us create the perfect team for your project

🕒 Your Available Times

Monday	9-11am	2-4pm	7-9pm
Tuesday	9-11am	2-4pm	7-9pm
Wednesday	9-11am	2-4pm	7-9pm
Thursday	9-11am	2-4pm	7-9pm
Friday	9-11am	2-4pm	7-9pm

Your Skills

Writing

Slides

Coding

Research

Thursday

9-11am

2-4pm

7-9pm

Friday

9-11am

2-4pm

7-9pm

Your Skills

Writing

Slides

Coding

Research

Your Commitment Level

Low

Medium

High

Willing to be a leader?

Take on coordination and decision-making responsibilities

Generate My Team

Management Prototype: Iteration

This week, I will introduce the figma page to both my instructor and classmates, asking them for their feedback on this page, including any issues they notice and suggestions for improvement.

Management Prototype: Iteration

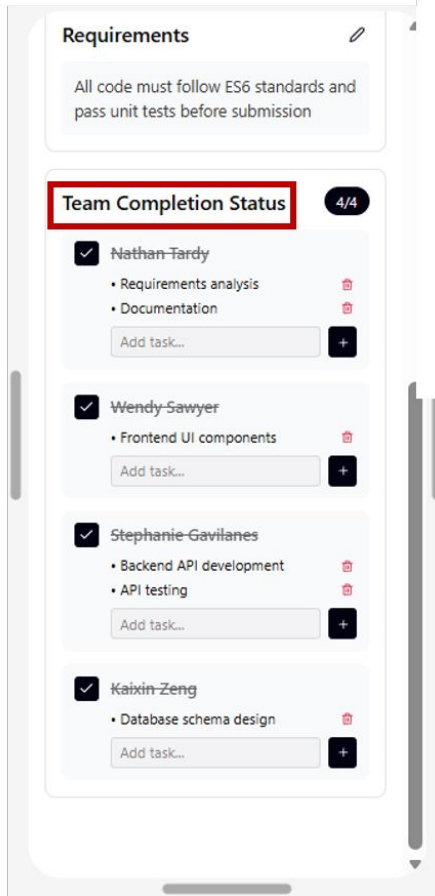
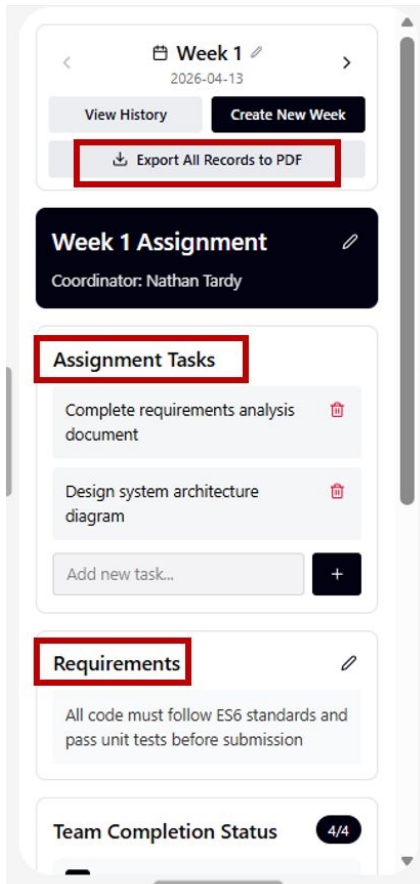
Iteration 1

The left screen of the mobile app prototype is titled "Align" with the subtitle "Build Efficient Teams". It features a purple "Add New Member" button at the top. Below it is a form for adding a member with a "Name" label and an input field containing the placeholder text "Enter member name". Further down is a section titled "My Skills" (highlighted with a red box) which includes a text input for "Skill name (e.g. Python, Design, Project)" and a "Proficiency Level: 3/5" slider. A blue "+ Add Skill" button is located below the skill input. At the bottom of this screen is a "Weekly Availability (10:00 AM - 6:00 PM)" section (also highlighted with a red box), which contains a grid for selecting availability across days of the week (Mon, Tue, Wed, Thu, Fri) and time slots (10:00, 10:30, 11:00, 11:30, 12:00, 12:30, 13:00, 13:30, 14:00). The bottom navigation bar includes icons for "Members", "Match", and "History".

The right screen of the mobile app prototype is also titled "Align" with the subtitle "Build Efficient Teams". It features a "Smart Group Matching" section (highlighted with a red box) which states: "System will automatically match optimal groups based on skills and availability". Below this is a "Group Size: 3 members" slider, with "2 members" on the left and "6 members" on the right. A "Match Information" box shows "Current members: 0" and "Expected groups: 0". A grey "Start Smart Matching" button is positioned below the match information. At the bottom of this screen, a red error message reads: "Not enough members. Please add more members first." The bottom navigation bar includes icons for "Profile", "Members", "Match", and "History".

1. **Team-matching** tool is not enough for **long-term support**
2. Misalignment can still happen during **actual assignment process**
3. Teams **do not need** to work at the same time , as long as tasks are completed before deadlines

Final Product



Weekly Assignment Records

Week 1

Date: 2026-04-13
Coordinator: Nathan Tardy

Assignment Tasks:

- Complete requirements analysis document
- Design system architecture diagram

Requirements:

All code must follow ES6 standards and pass unit tests before submission

Team Members:

- ☐ Nathan Tardy
 - Requirements analysis
 - Documentation
- ☐ Wendy Sawyer
 - Frontend UI components
- ☐ Stephanie Gavilanes
 - Backend API development
 - API testing
- ☐ Kaixin Zeng
 - Database schema design

- ☐ Wendy Sawyer
 - Frontend UI components
- ☐ Stephanie Gavilanes
 - Backend API development
 - API testing
- ☐ Kaixin Zeng
 - Database schema design

Week 2

Date: 2026-04-20
Coordinator: Wendy Sawyer

Assignment Tasks:

- Implement user login module
- Complete database design

Requirements:

Code must have detailed comments and all APIs need error handling

Team Members:

- ☐ Nathan Tardy
 - User authentication module

1. Helps teams build a **shared understanding** before dividing tasks
2. Increases **transparency and accountability**
3. Supports **progress tracking** and **fairer instructor evaluation**